Dr. Haytham Ashraf Abdelaal Ali

JSPS Postdoctoral Research Fellow – University of Tsukuba, Japan
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Research Interests

My research focuses on artificial intelligence for medical imaging and multimodal clinical intelligence. I develop deep learning and generative modeling approaches for computational imaging and inverse problems, integrating physics-based modeling with data-driven learning to build robust and interpretable AI systems for volumetric medical data. I am particularly interested in multimodal learning frameworks that combine medical imaging with clinical and contextual data to support reliable 3D image reconstruction, segmentation, and clinically deployable decision-support systems. My work emphasizes model robustness, uncertainty awareness, and translational impact in real-world clinical environments.

Keywords: Medical imaging AI, Generative models, Multimodal learning, Computational imaging, Inverse problems, Clinical decision support.

Work Experience

JSPS Postdoctoral Research Fellow 07/2024–Present
Systems and Information Engineering
University of Tsukuba, Japan

Lecturer 03/2024– 06/2024
Mathematics Department, Faculty of Science
Sohag University, Egypt

Assistant Lecturer 12/2023– 03/2024
Mathematics Department, Faculty of Science
Sohag University, Egypt

Teaching Assistant 07/2021–03/2023
Systems and Information Engineering
University of Tsukuba, Japan

Assistant Lecturer 11/2018– 09/2020
Mathematics Department, Faculty of Science
Sohag University, Egypt

Research Assistant 10/2014– 10/2018
Mathematics Department, Faculty of Science
Sohag University, Egypt

Education

PhD in Engineering 10/2020–09/2023
Systems and Information Engineering, University of Tsukuba, Japan
Thesis: *Parametric Level-Set Method for Geometric Tomography Reconstruction*

Master of Science in Mathematics 04/2016–09/2018
Faculty of Science, Sohag University, Egypt

Bachelor of Science 09/2009–07/2013
Graduated with Distinction (Excellent with Honors)
Faculty of Science, Sohag University, Egypt

Technical Skills

- **Deep Learning & AI:** PyTorch; generative models, attention-based architectures, and deep learning methods for medical imaging and inverse problems.
- **Medical Imaging & Computational Imaging:** CT reconstruction (FBP and iterative methods), forward/adjoint modeling, sparse-view reconstruction; ASTRA Toolbox, Torch Radon.
- **Programming:** Python, MATLAB, Mathematica.
- **Scientific Computing & Data Analysis:** NumPy, SciPy, Matplotlib, Jupyter.

Grants & Awards

Competitive Fellowships & Major Grants

- **JSPS Research Fellowship**,Prestigious fellowship awarded by the Japan Society for the Promotion of Science (JSPS) for research excellence 07/2024–07/2026
- **Grant-in-Aid for JSPS Fellows**, (KAKENHI Project Number: 24KF0101), University of Tsukuba, Japan, Co-Investigator 07/2024–03/2027
- **NEC C&C Foundation Research Grant**, University of Tsukuba, Japan, Principal Investigator 10/2022–03/2023
- **Monbukagakusho (MEXT) Scholarship**, Ministry of Education, Culture, Sports, Science and Technology (MEXT), Japan 10/2020–09/2023
- **Science & Technology Development Fund (STDF)**, Sohag University, Egypt, Co-Principal Investigator 04/2020–03/2021

Awards & Honors

- **Best Presentation Award**, International Conference on Computer and Automation Engineering (ICCAE), Perth, Australia, 2025
- **Excellent Oral Presentation Award**, ICCAE, Sydney, Australia, 2023
- **Academic Group Chair Award**, for outstanding research achievements during my Ph.D., University of Tsukuba, Japan, 2023

Travel & Other Grants

- **Travel Grant**, RIMS Workshop on Inverse Problems, Medical Imaging, and Related Topics, Kyoto, Japan, 2023

Selected Publications

Refereed Journal Articles

1. PIDA-GAN: Physics-Informed Dual-Stage Attention GAN with Data Consistency for Sparse-View CT Reconstruction
   Haytham A. Ali, Hiroyuki Kudo
   Under Review, 2026.

2. Compressed Sensing-Based Image Reconstruction for Discrete Tomography with Sparse View and Limited Angle Geometries
   Haytham A. Ali, Essam Rashed, Hiroyuki Kudo
   PLOS ONE, 2025.

3. Binary Tomography Reconstruction with Limited-Data by a Convex Level-Set Method
   Haytham A. Ali, Hiroyuki Kudo
   CMC-Computers Materials & Continua, 2022.

4. Generating Bézier curves for medical image reconstruction
   H. S. Abdel-Aziz, E. A. Zanaty, **Haytham A. Ali**, and M. K. Saad
   Results in physics, 2021.

Refereed Conference Papers

1. **Haytham A. Ali**, Hiroyuki Kudo, PCDS-GAN: Physics-Constrained Dual-Stage Generative Model with Adaptive Gating for Ultra Sparse-View CT Reconstruction, **ISBI** 2026 (Accepted).

2. **Haytham A. Ali**, Hiroyuki Kudo, Multi-Frequency Domains with Modified U-Net for Limited-Angle CT, **ISBI** 2026 (Accepted).

3. **Haytham A. Ali**, Hiroyuki Kudo, FLoAT-GAN: Frequency- and Locality-Guided Attention for Limited-Angle CT Reconstruction, **AAAI-26** W3PHIAI Workshop (Long Presentation), 2026.

4. **Haytham A. Ali**, Hiroyuki Kudo, Attention-Guided Frequency and Locality Learning GAN for LACT, **IWAIT/IFMIA**, 2026.

5. **Haytham A. Ali**, Hiroyuki Kudo, An Attention-Based Generative Adversarial Network for Limited-Angle CT Reconstruction, **ICCAE**, 2025.

6. **Haytham A. Ali**, Hiroyuki Kudo, Level-Set Method for Limited-Data Reconstruction in CT using Dictionary-Based Compressed Sensing, **ICCAE**, 2023.

7. **Haytham A. Ali**, Hiroyuki Kudo, New Level-Set-Based Shape Recovery Method and its application to sparse-view shape tomography, **DMIP**, 2021.

Presentations & Invited Talks

1. **Haytham A. Ali**, Hiroyuki Kudo, Novel Dual-Stage GAN for High-Fidelity CT Image Reconstruction from Sparse view CT, Annual Meeting of the Japanese Society for Medical Imaging (JAMIT), 2025.

2. **Haytham A. Ali**, Hiroyuki Kudo, Self-Attention GAN for High-Quality Limited-Angle CT Reconstruction, Annual Meeting of the Japanese Society for Medical Imaging (JAMIT), 2025.

3. **Haytham A. Ali**, Hiroyuki Kudo, Towards Accurate Reconstruction from Sparse and Limited-Angle CT Data Using Deep Learning, Asia Symposium on Image Processing, 2025.

4. **Haytham A. Ali**, Hiroyuki Kudo, Parametric Level Set Methods for Limited Data CT Image Reconstruction, Workshop on Inverse Problems, Medical Imaging, and Related Topics, 2023.

5. **Haytham A. Ali**, Katsuya Fujii, Hiroyuki Kudo, Reconstruction method for binary limited-data tomography using a dictionary-based sparse shape recovery, Annual Meeting of the Japanese Society for Medical Imaging (JAMIT), 2022.

Teaching Experience

Lecturer, Sohag University 03/2024–06/2024
Courses: Calculus, Linear Algebra, Geometry, and Double integrals

Responsibilities: delivery of lectures and tutorials, preparation of course materials and assessments, grading examinations, and providing academic advising and office hours.

Teaching Assistant, Sohag University 12/2023–03/2024
Courses: *Geometry, Differential Geometry, Calculus, Integral Calculus, and Linear Algebra*
Responsibilities: preparation of lecture materials, grading assignments and exams, and guiding students during tutorials and office hours.

Teaching Assistant, University of Tsukuba 07/2021–03/2023
ALD503: Computer Science
ALD504: Research in Computer Science
Responsibilities: organizing and preparing seminars, monitoring attendance, and mentoring students on course activities.

Teaching Assistant / Instructor, Sohag University 04/2016–09/2020
Courses: *Analytic Geometry, Differential Geometry, Calculus, Integral Calculus, Linear Algebra, Statics, and Dynamics*
Responsibilities: preparation of lecture materials, grading assignments and exams, guiding students during tutorials and office hours, and solving exercises with students.

References

- **Prof. Hiroyuki Kudo** 10/2020 – Present
  Ph.D. Advisor & Host, University of Tsukuba
  `kudo@cs.tsukuba.ac.jp`
- **Prof. Essam Rashed** 10/2020 – Present
  Collaborator & Mentor, University of Hyogo
  `rashed@gsis.u-hyogo.ac.jp`
- **Prof. M. Khalifa Saad** 04/2016 – 10/2020
  Advisor, Sohag University
  `mohamed.khalifa77@science.sohag.edu.eg`